# MILITARY SPECIFICATION

## FILTER, PARTICULATE, 2500 CFM, C30

### 1. SCOPE

1.1 This specification covers a particulate filter for use in removing aerosols from the air forced or drawn through the filter.

### 2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiber.

MILITARY

MIL-L-10547 - Liners, Case, Waterproof.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-282 - Filter Units, Protective Clothing, Gas Mask Components and Related Products; Performance Test Methods.



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DRAWINGS

CHEMICAL CORPS

C5-19-2128 - Filter, Particulate, 2500 CFM, C30 - Bill of Material.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing C5-19-2128 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the item shall conform to the specifications and drawings listed on Drawing C5-19-2128 and subsidiary drawings.

3.2 Manufacture and assembly. The filter shall be manufactured, assembled and marked as shown on Drawing C5-19-2115 and subsidiary drawings, which are listed on Drawing C5-19-2128.

3.3 Air resistance. The air resistance of the painted filter measured as a pressure differential at an airflow rate of 1000 cubic feet per minute (c.f.m.) shall be not greater than 0.50 inches of water when tested as specified in 4.5.1.

3.4 DOP smoke penetration. The filter shall show a DOP smoke penetration of not more than 0.02 percent when tested as specified in 4.5.1.

3.5 Resistance to rough handling. The filter shall comply with the air resistance and DOP smoke penetration requirements (see 3.3 and 3.4) before and after being subjected to the rough handling test as specified in 4.5.2.

3.6 Workmanship. The filter shall be free from contamination (foreign matter), defects, or damage that might impair its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier's responsibility. Unless otherwise specified herein, the suprlier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as



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otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all materials and components are in accordance with 3.1 and section 5 (see 6.3).

4.3 Alternate inspection (including testing) procedures. The supplier may utilize any alternate inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 Lotting. A lot shall consist of the filters produced by one manufacturer from the same materials and under essentially the same manufacturing conditions. However, not more than two lots of filter material shall be represented in any one lot of filters.

4.4.2 Sampling.

4.4.2.1 Nondestructive examination. Sampling for nondestructive inspection shall be comducted in accordance with Standard MIL-STD-105. The level and degree of inspection shall be specified in the contract.

4.4.2.2 Rough handling. Eighteen filters, 6 for test and 12 for retest, shall be taken at random from each regular production lot for the rough handling test.

4.4.3 Examination. The regular production lot shall be inspected in accordance with the classification of defects and Standard MIL-STD-105.

4.4.4 Rough handling. After subjecting assembled filters to the DOP penetration and air resistance test (4.5.1), the samples taken as specified in 4.4.2.2 shall be tested in accordance with 4.5.2.



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4.4.5 Classification of defects.

4.4.5.1 Filter, Particulate, 2500 c.f.m., C30 - assembly. (Dwg. C5-19-2115)

| Categories and defects | Acceptance standards |
|---|---|
| **Critical** | |
| *A. DOP penetration | 4.5.1 |
| *B. Air resistance | 4.5.1 |
| **Major** | AQL 1.0 percent defective |
| 1. Component missing or incorrectly assembled | |
| 2. Damage affecting functioning | |
| 3. Adhesive material incorrectly applied | |
| 4. Marking incorrect | |
| **Minor** | AQL 2.5 percent defective |
| 51. Contamination | |
| 52. Marking missing or illegible | |

*Each filter in the lot shall be inspected for DOP penetration and air resistance.

4.4.5.2 Preparation for delivery. (section 5)

| Categories and defects | |
|---|---|
| **Major** | AQL 4.0 percent defective |
| 1. Fiberboard sheets missing from filter faces | |
| 2. Containers incorrect | |
| 3. Contents missing from shipping container | |
| 4. Container closure incorrect | |
| 5. Case liner missing or incorrectly closed | |
| 6. Container or case liner damaged | |
| 7. Strapping loose or missing | |
| 8. Marking incorrect, missing or illegible | |
| **Minor** | AQL 6.5 percent defective |
| *51. Filler pads missing or insufficient to make tight pack | |

*Check prior to closing box.

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4.5 Tests.

4.5.1 Air resistance and DOP penetration. The air resistance and DOP penetration shall be determined in accordance with Standard MIL-STD-282, method 102.9.1, using an airflow of 1000 c.f.m.

4.5.2 Rough handling resistance. The samples selected shall be tested in accordance with Standard MIL-STD-282, method 105.9. The filters shall be mounted so that the fluted separators are in a vertical position. The duration of the vibrational period shall be a minimum of 15 minutes. After being rough handled, the filters shall again be subjected to the test specified in 4.5.1.

4.6 Acceptance/rejection criteria. If one of the filters taken for test fails when tested as specified in 4.5.2, the filters taken for retest shall be tested. If more than one filter fails in the test and retest combined, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1. Packing.

5.1.1 Level A. One filter with its inlet and outlet openings protected by 24 by 46-1/2-inch sheets of W5c fiberboard, conforming to Specification PPP-B-636, shall be packed in a nailed wood box conforming to class 2, style 4 or 4-1/2 of Specification PPP-B-621. Prior to packing in box the filters shall be placed within a fitted waterproof case liner conforming to type I, grade B, class 2 of Specification MIL-L-10547, with closure to be in accordance with the appendix thereto. Filler pads fabricated from material conforming to Specification PPP-B-636 shall be placed between the top, bottom, sides and ends of the wrapped filter and the box. Additional filler pads shall be used as required to make a tight pack. The box shall be closed in accordance with Specification PPP-B-621 and strapped in accordance with the appendix thereto. Galvanized strapping shall be used.

5.1.2 Level B. Not applicable.

5.1.3 Level C. The filter shall be packed as specified in 5.1.1 except that a class 1 nailed wood box and coated finish strapping shall be used.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The filter covered by this specification is a component of Filter, Gas and Particulate, 2500 CFM, M16, which is used for protecting fixed installations against toxic gases and aerosols.



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6.2 Ordering data. Procurement documents should specify:

(a) The title, number, and date of this specification.
(b) The level of packing required.

6.3 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

6.4 Pilot lot. A pilot lot may be required to evaluate the supplier's production and quality control systems. The contracting officer will specify whether a pilot lot is required and applicable pilot lot acceptance criteria.

6.5 Manufacturing procedure. Filters manufactured in the following manner have been found to satisfy the performance requirements of this specification:

(a) Pleat filter material conforming to Specification MIL-F-13785 on a suitable filter pleating machine using fiberboard separators to form and hold the pleats. Cut off a section of pleated material containing 150 folds (75 folds per side) or the number of folds which have been found to give the desired compactness. Place this section on a temporary U frame approximately 22-1/2 inches in length by 46-1/2 inches wide with ends 10-3/4 inches high inside dimensions. Remove the exposed fiberboard separators and insert in their place fluted kraft paper separators which are approximately 22-1/2 inches long and have been fluted on the day of assembly. Repair any torn places in the filter material with adhesive for filters. Turn the pleated assembly over and repeat the operation.

(b) Check the pleated section for proper compactness. Excessively tight packing will cause the fluted separators to cut into the filter material, and excessively loose packing will allow undesirable movement of the separators. Correct compactness may be obtained by removing or splicing in sections of pleated filter material to the ends of the assembly.

(c) Insert fiberboard backing strips between each end of the pleated assembly and the temporary frame to keep pleats from banding over when trimmed. Trim both sides of pleated assembly with a high speed band saw to obtain correct width. It is necessary to have an angle of not less than 30 degrees between the saw blade and filter pleats when trimming to prevent the saw from tearing the paper.

(d) Apply one coat of adhesive for filters to both sides of the pleated assembly. Allow to penetrate and dry approximately 2 hours. Apply second coat of adhesive. Allow this coat to penetrate and dry approximately 4 hours. Apply third coat of adhesive and allow to penetrate and dry approximately 4 hours. Apply additional coats if necessary to cover any pin holes



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which might arise when solvent evaporates. After careful inspection to see that there are no pin holes remaining, trim excessive adhesive from edges of assembly and remove temporary frame.

(e) Assemble the inner and outer panels around the pleated assembly to form a permanent frame. This frame should be flush to one edge of the pleated assembly. Apply one coat of adhesive around entire periphery, sealing the frame to the filter. Allow this adhesive to penetrate and dry approximately 2 hours. Apply second coat of adhesive and allow it to penetrate and dry approximately 4 hours. Turn filter over and repeat cementing operation on reverse side. Fasten plywood or fiberboard cover on flush side of filter to prevent damage during further handling. Fill all holes in edges of the frame with felt and then coat the edges with adhesive.

(f) Perform final test for DOP penetration and air resistance. After testing, mark filter.

6.6 Supersession data. This specification supersedes Chemical Corps Purchase Description 197-54-584A, dated 19 November 1958.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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